

**SET-4****Series %BAB%/C****प्रश्न-पत्र कोड
Q.P. Code 56/B/6**

रोल नं.

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Roll No.

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परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.

- कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 11 हैं।
- प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।
- कृपया जाँच कर लें कि इस प्रश्न-पत्र में 12 प्रश्न हैं।
- कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें।
- इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
- Please check that this question paper contains 11 printed pages.
- Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains 12 questions.
- **Please write down the serial number of the question in the answer-book before attempting it.**
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.



रसायन विज्ञान
(केवल दृष्टिबाधित परीक्षार्थियों के लिए)

**CHEMISTRY****(FOR VISUALLY IMPAIRED CANDIDATES ONLY)**

निर्धारित समय : 2 घण्टे

Time allowed : 2 hours

अधिकतम अंक : 35

Maximum Marks : 35



General Instructions :

Read the following instructions very carefully and strictly follow them :

- (i) *This question paper contains **12** questions. **All** questions are compulsory.*
- (ii) *This question paper is divided into **three** Sections – **Section A, B and C.***
- (iii) ***Section A** – Questions no. **1** to **3** are very short answer type questions, carrying **2** marks each.*
- (iv) ***Section B** – Questions no. **4** to **11** are short answer type questions, carrying **3** marks each.*
- (v) ***Section C** – Question no. **12** is a case based question, carrying **5** marks.*
- (vi) *Use of log tables and calculators is **not** allowed.*

SECTION A

1. Arrange the following compounds in the increasing order of their property indicated : (Any **two**) $2 \times 1 = 2$
 - (i) Aniline, $\text{CH}_3 - \text{NH}_2$, p-nitroaniline (Basic strength)
 - (ii) $\text{C}_6\text{H}_5\text{NH}_2$, $\text{C}_6\text{H}_5\text{NHCH}_3$, $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$ (Basic strength)
 - (iii) $\text{C}_2\text{H}_5\text{NH}_2$, $(\text{C}_2\text{H}_5)_2\text{NH}$, $\text{C}_6\text{H}_5\text{NH}_2$ (Solubility in water)
2. Write two differences between Order of reaction and Molecularity of reaction. **2**
3. Write the chemical equations involved in the following reactions : $2 \times 1 = 2$
 - (i) Clemmensen reduction
 - (ii) Rosenmund reduction



SECTION B

4. Account for the following : 3×1=3

- (i) Transition metals and their compounds show catalytic activities.
- (ii) Cu^{2+} is coloured while Cu^+ is colourless in aqueous solution.
- (iii) E^0 value for Mn^{2+}/Mn couple is highly negative.

5. (a) The rate constant for a first order reaction is 40 s^{-1} . How much time will it take to reduce the initial concentration of the reactant to its $\frac{1}{10}$ th value ? $[\log 10 = 1]$ 3

OR

(b) The reaction between A and B is first order with respect to A and zero order with respect to B. For this reaction, fill in the blanks in the following table : 3

Experiment	[A]/M	[B]/M	Initial rate M/min
I	0.1	0.1	2.0×10^{-2}
II	—	0.2	4.0×10^{-2}
III	0.4	0.4	—
IV	—	0.2	2.0×10^{-2}

6. (a) Define the following terms with a suitable example in each : 3×1=3

- (i) Multimolecular colloid
- (ii) Sol
- (iii) Adsorption

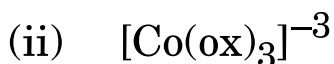
OR

- (b) (i) Write the Freundlich adsorption isotherm at (I) low pressure, and (II) high pressure.
- (ii) Why is adsorption exothermic in nature ? What is the effect of increasing temperature on adsorption ? 3



7. Define transition elements. Why do transition metals have higher enthalpies of atomisation ? Which element in 3d series has lowest enthalpy of atomisation and why ? 3

8. (a) Write the hybridization and magnetic character of the following complexes on the basis of valence bond theory :



[Atomic number : Fe = 26, Co = 27]

(b) Write the formula of the following complex :

Hexaamminecobalt(III)chloride 2+1=3

9. (a) Write the reactions of ethanal with 3×1=3



OR

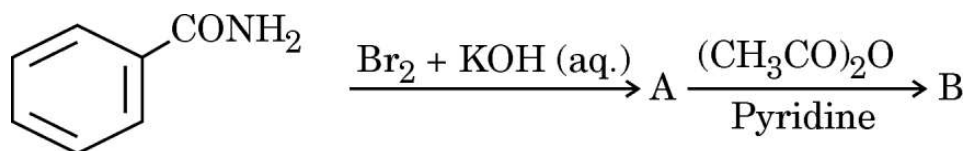
(b) How will you convert the following : 3×1=3

(i) Ethanoic acid to 2-chloroethanoic acid

(ii) Propanone to t-butyl alcohol

(iii) $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_2 - \text{CN}$ to $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_2 - \text{CHO}$

10. (a) (i) Why does aniline not give Friedel-Crafts reaction ?
(ii) Write the equation involved in carbylamine reaction.
(iii) Write the structures of A and B in the following :



3×1=3

OR

(b) Describe Hinsberg method for the identification of primary, secondary and tertiary amines. 3



11. (a) Arrange the following compounds in increasing order of their reactivity in nucleophilic addition reactions :
Ethanal, Propanal, Propanone, Butanone
- (b) Why does Benzaldehyde not undergo aldol condensation ?
- (c) Write a simple chemical test to distinguish between Pentan-2-one and Pentan-3-one. 3×1=3

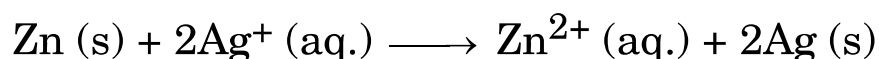
SECTION C

12. Read the passage given below and answer the questions that follow :

An electrochemical cell consists of two metallic electrodes dipping in electrolytic solutions. Electrochemical cells are of two types. In galvanic cell, the chemical energy of spontaneous redox reaction is converted into electrical work, whereas in an electrolytic cell, electrical energy is used to carry out a non-spontaneous redox reaction. The standard electrode potential for any electrode dipping in an appropriate solution is defined with respect to standard electrode potential of hydrogen electrode taken as zero.

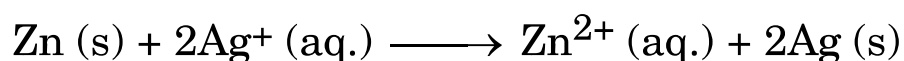
The conductivity (k) of an electrolytic solution depends on the concentration of the electrolyte, nature of solvent and temperature. Both conductivity and molar conductivity (Λ_m) vary with concentration of solution.

- (i) Depict the galvanic cell in which the following reaction takes place :





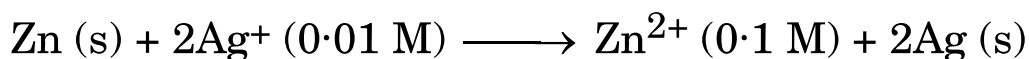
- (ii) What happens when external potential applied is greater than E_{cell}^0 of electrochemical cell ?
- (iii) How do conductivity and molar conductivity vary with concentration ?
- (iv) (a) Calculate $\Delta_r G^0$ of the following reaction :



$$\text{Given : } E_{\text{cell}}^0 = 1.56 \text{ V, } 1 \text{ F} = 96500 \text{ C mol}^{-1}$$

OR

- (b) Calculate the emf of the following cell at 25°C :



$$\text{Given : } E_{\text{cell}}^0 = 1.56 \text{ V}$$

$$[\log 2 = 0.3010, \log 3 = 0.4771, \log 10 = 1]$$

$$1+1+1+2=5$$